MILITARY SPECIFICATION

CONSOLE, SHIPS AIRCRAFT INERTIAL SYSTEMS ALIGNMENT (SAISAC)

1. SCOPE

1.1 This specification covers the design of the shipboard equipment required for erection and alignment of aircraft inertial navigational systems at sea from shipboard and aircraft sources.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-E-1 - Electron Tubes and Crystal Rectifiers.
MIL-S-901 - Shockproof Equipment, Class HI (High Impact) Shipboard Application; Tests for.
MIL-I-983 - Interior Communication Equipment, Naval Shipboard; Basic Design Requirements for.
MIL-C-5015 - Connectors, Electric "AN" Type.
MIL-A-15344 - Amplifier, Synchro Signal.
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts; Preparation for Delivery of.

STANDARDS

MILITARY

MIL-STD-200 - Electron Tubes; Selection and Use of.
MIL-STD-701 - Preferred and Guidance List of Semiconductor Devices.

FSC 6605

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DRAWINGS

BUREAU OF SHIPS
SKN-665H-62-1 - Schematic Wiring Diagram - Ships Aircraft Inertial System Alignment Console (SAISAC)

(Copies of specifications, standards, drawings, and publications required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Applications for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N.Y.)

3. REQUIREMENTS

3.1 Preproduction sample. - A preproduction sample shall be furnished for the tests specified in 4.3.

3.2 Description. - The Ships Aircraft Inertial Systems Alignment Console (SAISAC) provides continuous alignment information from shipboard sources in the form of synchro signals for aligning inertial autonavigators mounted in aircraft on board. The console shall contain the servo amplifiers, synchro components and switching necessary to provide 400 cycle synchro signals of own ship's course at 360 degrees per revolution and north-south and east-west components of own ship's speed at 100 knots per revolution with sufficient capacity to erect and align thirty aircraft autonavigators simultaneously.

3.3 Operation. - The SAISAC shall be capable of receiving inputs from several sources of own ship's course and north and east velocity components of own ship's motion as indicated on Bureau of Ships Drawing SKN-665H-62-1. Amplifiers shall be provided where required for signal strength or reference power compatibility. These inputs shall be taken to appropriate selector switches. One selector switch shall provide outputs of own ship's heading and velocity components from various shipboard sources for master aircraft velocity reference and for heading alignment of all aircraft inertial autonavigators. A second selector switch shall provide velocity components

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from various shipboard sources or from the master aircraft inertial auto-navigator for alignment of aircraft inertial navigators. Damping control for varying degrees of ship movement shall be provided.

3.4 General features. - The equipment shall be in accordance with the following paragraphs of MIL-I-983 in addition to the requirements specified herein. (Whenever a requirement of MIL-I-983 conflicts with a requirement of this specification, the requirement of this specification shall govern.)

General information -
Definitions -
Materials, general -
Stressed or vital parts -
Substitution of (equal or superior) materials-
Fungus-inert materials-
Unacceptable materials-
Acceptable materials-
Flammable materials -
Arc-resistant materials -
Toxic materials -
Wood -
Aluminum-
Magnesium -
Iron or steel -
Brass -
Bronze -
Copper -
Copper-beryllium alloy -
Copper-nickel alloy -
Nickel-copper alloy (Monel) -
Copper-nickel-zinc alloy (nickel silver and German silver) -
Zinc -
Other metals -
Springs -
Plastics -
Ceramics -
Impregnating, filling, potting, embedding and encapsulating compounds
Glass -
Lubricants and lubrication -

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Painting -
Protection against corrosion -
Bolts, machines screws, studs and nuts -
Gaskets -
Dials and pointers -
Dial sizes -
Locking devices -
Washers -
Ball bearings -
Parts-electrical-general-
Electron tube or capacitor sockets -
Capacitors -
Variable resistors -
Transformers -
Relays -
Synchros (see 3.7.1) -
Electrical tapes -
Batteries -
Dial illumination lamps -
Switches -
Indicator lights and lampholders -
Fuses -
Fuseholders and fuseclips -
Metallic rectifiers -
Enclosures-general -
Enclosure-accessibility -
Enclosure-degree of (see 3.6.1) -
Enclosure-mounting -
Stiffening grooves -
Minimum sheet metal thickness -
Through bolting -
Stuffing tubes -
Ventilation -
Threaded devices -
Rounded corners and edges -
Internal subassembly protection -
Drilled and tapped holes -
Welding -
Temperature and humidity -
Accelerated life -
Shock, vibration and inclination (see 3.7.6) -
Primary power supply circuits -

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Power supply tolerances (Supply line voltage and frequency variation)
Personnel protection -
Shielding and radio frequency noise reduction -
Ground potential and grounding -
Soldering -
Electrical parts mounting -
Internal subassembly connection (test cable required) -
Terminal boards, connectors and terminals -
Synchro connections and markings -
Wiring -
Color coding -
Dial illumination -
Electrical insulation -
Dielectric strength and insulation resistance clearances.-
Airborne audiable noise -
Drawings - general -
Drawings - preliminary -
Drawings - working -
Schematic diagrams -
Wiring diagram -
Drawing list -
Assembly drawings -
List of onboard repair parts -
Drawings - manufacturing -
Bill of materials -
Interchangeability and standardization (see 3.5.9) -
Technical manuals (service manuals) -
Repair parts -
Repair parts (electronic) -
Designation and marking (serial numbers are required) -
Design for production -
Reports -
Nomenclature -
Workmanship and general examination -

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3.5 Design. - The SAISAC shall be so designed that vibration, shock, or acceleration, which may be encountered in service aboard any of the type of ships for which the equipment is intended, will not adversely affect the accuracy of the equipment. The equipment shall be so designed that the operator will not receive an electric shock when making any ordinary adjustment while the equipment is in operation.

3.5.1 Excessive play between moving parts, resulting in noise when the equipment is subject to vibration, shall be eliminated insofar as practicable.

3.5.2 The equipment and its operation shall not be adversely affected by magnetic fields up to 5 gausses.

3.5.3 Adjusting devices shall be free from excessive friction. Each adjusting device requiring a setting, shall be provided with a suitable graduate dial, scale, or indicator, which shall be so designed that it may be read without difficulty. Handles of adjusting devices shall be so constructed or protected that accidental contacts will not disturb them or change their setting, or shall be provided with protective covers attached to the unit in such a manner as not to cause interference or affect ease of handling.

3.5.4 Mechanical components shall be of a suitable material, properly and evenly hardened or treated to prevent wear. They shall, wherever practicable, be provided with oil-saturated felt wipers for lubrication. Gear trains shall be designed for minimum backlash consistent with good response.

3.5.5 Provision shall be made for adequate lubrication of all bearings.

3.5.6 The amplifiers required in the SAISAC shall comply with all the requirements of MIL-A-15344 for similar equipment. The amplifiers provided for velocity components of own ships motion shall be interchangeable and shall be designed so that input CT's may be by-passed and the amplifier accept as inputs the signals transmitted from a synchro resolver (Type 23RS4 or equivalent). These outputs of the synchro resolver are 400 cycle signals with voltage proportional to velocity components.

3.5.7 Electrical connections from each unit to external circuits shall be brought out to common AN termination connectors in accordance with type 3SWA cable of MIL-C-5015. Both plug and jack shall be provided by the contractor.

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3.5.8 Parts of the equipment requiring accurate placing shall be marked or doweled, so that when disassembled for repairs or adjustments, the parts may be replaced in their proper position. Parts requiring occasional renewal due to unavoidable service wear shall be designed for maximum practicable ease of replacement and alignment.

3.5.9 Interchangeability. - Except as specifically provided herein, similar parts and permanent assemblies, including repair parts or corresponding apparatus furnished on the same contract or order or built to the same drawings, shall be interchangeable without the necessity of further machining or hand fitting of any kind.

3.5.10 Proposed changes in design affecting interchangeability of parts or assemblies, including repair parts, shall be specifically approved as satisfactory to the bureau or agency concerned before being made effective.

3.5.11 Mercury, in any form, shall not be used in units of this system.

3.5.12 All 3SWA type cable shields of MIL-C-5015 shall be tied to a common point and grounded at that point within the equipment.

3.6 Mechanical features. -

3.6.1 Enclosures. - The SAISAC console shall be bulkhead mounted in a dripproof case (zero degree protection only). Provision shall be made to assure adequate natural ventialtion. Provision shall be made for the protection of gears from dust. Mounting and external wiring of the equipment shall be accomplished without removal of components from the enclosure, and without exposing the gears to dust. Internal subassemblies shall not be supported by the enclosure.

3.6.2 Accessibility. - All units and subassemblies of the equipment shall be easily accessible for repair and adjustment. Protection of the internal subassemblies shall be provided in accordance with MIL-I-983. Suitable supports shall be furnished where necessary to allow repair work without the units resting on electrical components, electrical connections, or on components that may be disarranged or otherwise damaged. Knurled knob captive screws with a screwdrive slot shall be used on all covers that require removal or opening for operating adjustments.

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3.6.3 Weight and size.- The weight and size of units of the SAISAC shall be kept to a minimum.

3.7 Electrical features.-

3.7.1 Amplifier units, synchro units, servo motors and synchro capacitors shall be in accordance with the requirements of MIL-A-15344.

3.7.2 Power supply.- The equipment shall operate from a power supply of single-phase, 400 cycle, 115 volts a.c. A twenty-eight volt direct current supply (rectifier) shall be provided in the SAISAC for the "on station" light and the damping circuit.

3.7.3 Synchronization, phase shift, and effect of synchro bus load on synchro transformers shall be in accordance with MIL-A-15344.

3.7.4 Synchro reference bus.- A synchro reference bus shall be provided in the SAISAC so that R_1 and R_2 connections can be made to all synchro transmitters which provide signals to the SAISAC directly and to the outputs of all amplifiers where synchro signals are amplified in the SAISAC. Provision shall also be made to provide reference power voltage outputs to aircraft which lag or lead the synchro reference power by 90 degrees.

3.7.5 Blown fuse indicators.- Blown fuse indicators shall be located on the front of the console to indicate loss of power supply and loss of excitation to each of the ten input circuits, on the input side of amplifier receiving excitation from the aircraft, and on the reference output circuits.

3.7.6 Shock and vibration.- The SAISAC shall conform to the HI (high impact) shock requirement of MIL-S-901. Equipment shall meet the vibration requirements of MIL-I-983.

3.7.7 Electron tubes.- All electron tubes shall be in accordance with MIL-E-1, and shall be selected from those types listed in MIL-STD-200. Tentative and final electron tube complements shall be listed on "Electron Tube Complement Report", from DD 816, and four copies of each form shall be forwarded to the bureau or agency concerned. (Copies of form DD 816 may be obtained from the Government inspector.)

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3.7.8 Semiconductor devices. - Semiconductor devices shall be selected from MIL-STD-701. The contractor shall submit a tentative semiconductor complement report to the bureau or agency concerned as soon as the equipment design requirements have been determined. In the event a change in the semiconductor complement is found necessary, it shall be reported within 30 days. In order to assure sufficient quantity for maintenance of equipment in the Naval service, the contractor shall furnish the bureau or agency concerned with a final semiconductor complement report after the equipment design has been completed, and before production has begun. These reports shall be submitted on the same form DD 816 as used for electron tubes, and shall be combined into a consolidated form for each equipment.

3.8 Controls and indicators. - The controls and indicators listed in this paragraph shall be located on the front panel of the SAISAC console.

3.8.1 Power switch. - A power switch and indicator light shall provide for switching on the 115-volt, 400-cycle power supplies. The switch shall break both sides of the circuit. A pilot light shall be provided to indicate that 115-volt, 400-cycle power is being supplied.

3.8.2 Master velocity reference and heading inputs selector switch. - A selector switch shall be provided for selection of any one of the shipboard (non-aircraft) sources of velcoity components and heading. The velocity component outputs of this switch shall provide alignment reference velocity to the master aircraft. The heading output shall provide heading to all aircraft connected to the SAISAC.

3.8.3 Alignment input selector switch. - A selector switch shall be provided for selection of any one of four sources of velcoity components for erection and alignment of any aircraft (less master) autonavigators connected to the SAISAC.

3.8.4 Aircraft "on station" lamp. - A lamp operating on 28 volt direct current (d.c.) shall be provided to indicate when one or more aircraft are connected to the SAISAC.

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3.8.5 Damping control selector switch. - A selector switch shall be provided which shall close appropriate relays in aircraft autonavigators operated on 28 volt d.c. This selector switch shall provide damping control positions:: "No damping", "K2 damping", "K3 damping", or "Hold Alignment".

3.8.6 Intercom connection. - A three conductor cable shall be connected through the SAISAC to provide for audio communication to aircraft over the SAISAC connection cables from a position in the vicinity of the SAISAC.

3.9 Accuracy. - The overall error of the SAISAC shall not exceed the values shown in MIL-A-15344 for similar equipment.

3.10 Commencement of work by manufacturer. - Upon receipt of drawing approval letter and prior to validation of drawings, and, if the manufacturer has no objections to any changes or modifications requested in the drawing approval letter, manufacture may be started in accordance with the drawings as revised in anticipation of validation of drawings.

3.11 Equipment performance. - Approval of drawings of equipment by the bureau or agency concerned or its field representatives does not relieve the manufacturer of the responsibility for satisfactory operation of the equipment under service conditions. If for any reason an equipment fails to meet in-service conditions, or specification requirements, the contractor shall correct any material defects or design deficiencies.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 The inspection procedures for the tests to be performed on the equipment shall be as specified in 4.3, 4.4, and 4.5. The attention of the manufacturer is invited to the applicable paragraphs of MIL-I-983 for information concerning these tests such as where conducted, failures, and test conditions.

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4.3 Preproduction inspection. - One sample of the SAISAC shall be subjected to the examination and tests specified in table I, which shall be performed in general in in the order listed.

Table I - Preproduction inspection.

Examination and tests	Requirement paragraph	Test paragraph
General examination	3.4	4.5.1
Accuracy	3.9	4.5.1
Supply line voltage and frequency variation	3.4	4.5.1
Shielding and radio frequency noise	3.4	4.5.1
Airborne audible noise	3.4	4.5.1
Temperature and humidity	3.4	4.5.1
Synchronization	3.7.3	4.5.1
Phase shift	3.7.3	4.5.1
Effect of magnetic fields	3.5.2	4.5.1
Accelerated life	3.4	4.5.1
Dielectric strength[1]	3.4	4.5.1
Insulation resistance[1]	3.4	4.5.1
Enclosure	3.4	4.5.1
Vibration	3.7.6.	4.5.1
Shock	3.7.6	4.5.1

[1]For all but rotating components of the equipment.

4.3.1 Acceptance of preproduction tests. - The results of each test shall be compared with the specification requirements. In the event of failure to conform to this specification, the contractor shall correct the cause of failure in the preproduction samples and on all future production units.

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4.4 Acceptance inspection. -

4.4.1 Group A and group B examination and tests. - Each equipment covered on the contract or order shall be subjected to the group A and group B examination and tests listed in table II with acceptance based on 4.4.3.

Table II - Acceptance inspection.

Examination and tests	Requirement paragraph	Test paragraph
Group A and B examination and tests		
General examination	3.4	4.5.1
Accuracy	3.9	4.5.1
Dielectric strength [1]	3.4	4.5.1
Insulation resistance [1]	3.4	4.5.1
Group C tests		
Supply line voltage and frequency variation	3.4	4.5.1
Synchronization	3.7.3	4.5.1
Phase shift	3.7.3	4.5.1
Effect of magnetic fields	3.5.2	4.5.1
Shielding and radio frequency noise	3.4	4.5.1
Temperature and humidity	3.4	4.5.1
Accelerated life	3.4	4.5.1
Enclosure	3.4	4.5.1
Vibration	3.7.6	4.5.1
Shock	3.7.6	4.5.1

[1] For all but rotating components of the equipment.

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4.4.2 Group C tests. - Group C tests shall be required by the Bureau of Ships when the basic design of the equipment or the material of a vital part has been changed. One complete equipment shall be selected and subjected to each of the group C tests listed in 4.1 with the acceptance based on 4.4.3.

4.4.3 Acceptance, groups A, B and C. - The results of each test shall be compared with specification requirements. In the event of failure to conform to this specification for any test, the contractor shall correct the cause of failure on future production units and repair the deficiency in all equipment produced on the contract or order.

4.5 Test procedures. -

4.5.1 Accuracy, synchronization, phase shift and effect of magnetic fields tests shall be conducted as specified in MIL-A-15344. Shock tests shall be conducted as specified in MIL-S-901. All other tests shall be conducted as specified in MIL-I-983.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of shipboard repair parts. -

5.1.1 Basic equipment. -

5.1.1.1 Preservation and packaging. - Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

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5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Shipboard repair parts.- The shipboard repair parts shall be preserved and packaged by level A; packed by level C and marked by levels A and C, respectively, in accordance with MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment.- The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.3).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made by a Government activity (see 6.3).

5.2.1.1 Preservation, packaging, packing, and marking.- The equipment and accessories, repair parts and technical publications shall be preserved and packaged by levels A or C; packed by levels A or B as specified (see 6.3), and marked in accordance with MIL-E-17555.)

6. NOTES

6.1 Intended use.- The equipment covered by this specification is intended for Naval service where it is expected to withstand continuous use for long periods, under Military service conditions without benefit of overhaul. The equipment is in each case a vital instrument intended for important use by the forces concerned. Failure at a critical moment invariably results in serious reduction in the battle efficiency of the ship.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Whether manufacturing drawings are required.
(c) Supply demand control point for repair parts (see 3.4).
(d) Quantity of technical manuals required.
(e) Preservation, packaging, packing or marking required, if other than as specified in paragraph 5.1 (see 5.2).

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Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy - Ships
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